# PRACTICAL 1A

**Aim: Write a Java Program to create simple calculator.**

**Input:**

```
import java.util.Scanner;

class Main {
  public static void main(String[] args) {

    char operator;
    Double number1, number2, result
    Scanner input = new Scanner(System.in)
    System.out.println("Choose an operator: +, -, *, or /");
    operator = input.next().charAt(0)
    System.out.println("Enter first number");
    number1 = input.nextDouble();

    System.out.println("Enter second number");
    number2 = input.nextDouble();

    switch (operator) {

      case '+':
        result = number1 + number2;
        System.out.println(number1 + " + " + number2 + " = " + result);
        break;

      case '-':
        result = number1 - number2;
        System.out.println(number1 + " - " + number2 + " = " + result);
        break;

      case '*':
        result = number1 * number2;
        System.out.println(number1 + " * " + number2 + " = " + result);
        break;
```

```
      case '/':
        result = number1 / number2;
        System.out.println(number1 + " / " + number2 + " = " + result);
        break;

      default:
        System.out.println("Invalid operator!");
        break;
    }

    input.close();
  }
}
```

**Output:**

```
Choose an operator: +, -, *, or /
+
Enter first number
1
Enter second number
2
1.0 + 2.0 = 3.0
```

# PRACTICAL 1B

**Aim: Write a Java Program to to make a class called as circle. It should have three methods namely: accept radius, calculate area and display the area.**

## Input:

```
import java.util.*;
class Circle {
   private float r, area;
   public void accept(float x) {
      r = x;
   }
   public void calculate() {
      area = 3.14f * r * r;
   }
   public void display() {
      System.out.println("Area = " + area);
  }
}
   public class Main {
   public static void main(String args[]) {
      float x;
   Scanner sc = new Scanner(System.in);
   System.out.print("Enter Radius: ");
      x = sc.nextFloat();
   Circle c = new Circle();
      c.accept(x);
      c.calculate();
      c.display();
   }
}
```

## Output:

```
Enter Radius: 6
Area = 113.04
```

# PRACTICAL 2A

**Aim: Write a Java program to demonstrate use of Access specifiers.**

**Input:**

```
import java.util.*;

class Employee {
   private int empid;
   private float salary;
   private String empname, designation;

   void get_employee() {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter employee name: ");
      empname = sc.nextLine();
      System.out.print("Enter designation: ");
      designation = sc.nextLine();
      System.out.print("Enter employee ID: ");
      empid = sc.nextInt();
      System.out.print("Enter salary: ");
      salary = sc.nextFloat();

      sc.close();
   }

   void show_grade() {
      if (salary < 10000)
         System.out.println("Grade D");
      else if (salary < 25000)
         System.out.println("Grade C");
      else if (salary < 50000)
         System.out.println("Grade B");
      else
         System.out.println("Grade A");
   }
```

```
    void show_employee() {
        System.out.println("\nEmployee Details:");
        System.out.println("Name: " + empname);
        System.out.println("Designation: " + designation);
        System.out.println("ID: " + empid);
        System.out.println("Salary: " + salary);
    }
}

public class Main {
    public static void main(String args[]) {
        Employee e = new Employee();
        e.get_employee();
        e.show_grade();
        e.show_employee();
    }
}
```

**Output:**

```
Enter employee name: Padmaj Jadhav
Enter designation: Data Scientist
Enter employee ID: 3532
Enter salary: 40000
Grade B

Employee Details:
Name: Padmaj Jadhav
Designation: Data Scientist
ID: 3532
Salary: 40000.0
```

# PRACTICAL 2B

**Aim: Write a Java program to demonstrate use of access specifiers.**

## 1.Date and Time:

## Input:

```
import java.util.*;
class DateTime
{
   public static void main(String args[])
   {
      Date d= new Date();
      System.out.println(d);
   }
}
```

## Output:

Fri Jul 11 21:54:19 GMT 2025

## 2.Calender:

## Input:

```
import java.util.*;
class DateTime
{
   public static void main(String args[])
   {
      String monthName[]= {"Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug",
"Sep", "Oct", "Nov", "Dec"};
      Calendar c= Calendar.getInstance();
      System.out.println("Date:"
+c.get(Calendar.DATE)+"/"+monthName[c.get(Calendar.MONTH)]+"/"+c.get(Calen
dar.YEAR));
      System.out.println("Time:"+c.get(Calendar.HOUR)+
":"+c.get(Calendar.MINUTE)+":"+c.get(Calendar.SECOND));
   }
}
```

## Output:

```
Date:11/Jul/2025
Time:9:56:18
```

# PRACTICAL 3A

**Aim: Write a Java Program to demonstrate ArithmeticException and NumberFormatException using user input in Java.**

**Input:**

```
import java.io.*;

class Divide {
   public static void main(String args[]) throws IOException {
      int a, b, res;
      BufferedReader br = new BufferedReader(new
InputStreamReader(System.in));
      String str;

      try {
         System.out.print("Enter the first number: ");
         str = br.readLine();
         a = Integer.parseInt(str);

         System.out.print("Enter the second number: ");
         str = br.readLine();
         b = Integer.parseInt(str);

         res = a / b;
         System.out.println("The Quotient = " + res);

      } catch (NumberFormatException e) {
         System.out.println("Error: Please enter valid integers.");
      } catch (ArithmeticException e) {
         System.out.println("Error: Division by zero is not allowed.");
      }
   }
}
```

## First Output:

Enter the first Number: 32
Enter the second Number: 35
The Quotient = 0

## Second Output:

Enter the first Number: 35
Enter the second Number: 0
ERROR!
Error: Division by zero is not allowed.

# PRACTICAL 3B

**Aim: Write a Java Program to perform division of two user-entered numbers and handle NumberFormatException and ArithmeticException using try-catch blocks.**

**Input:**

```
import java.io.*;

class Divide {
   public static void main(String args[]) throws IOException {
      int a = 0, b = 0, res;
      BufferedReader br = new BufferedReader(new
InputStreamReader(System.in));
      String str;

      try {
         System.out.print("Enter First Number: ");
         str = br.readLine();
         a = Integer.parseInt(str);

         System.out.print("Enter Second Number: ");
         str = br.readLine();
         b = Integer.parseInt(str);

         res = a / b;
         System.out.println("The Quotient = " + res);

      } catch (NumberFormatException ne) {
         System.out.println("Invalid Number. Please enter only integers.");
      } catch (ArithmeticException ae) {
         System.out.println("Exception occurred: Division by zero is not allowed.");
      }
   }
}
```

## First Output:

Enter the first Number: 32
Enter the second Number: 35
The Quotient = 0

## Second Output:

Enter the first Number: 35
Enter the second Number: 0
ERROR!
Error: Division by zero is not allowed.

## Third Output:

Enter First Number: 35
Enter Second Number: S
Invalid Number. Please enter only integers.

# PRACTICAL 3C

**Aim: Write a Java program to accept and display the month number. Throw a number format exception if improper month number is entered.**

## Input:

```
import java.io.*;
class month
{
   public static void main(String args[])throws IOException
   {
      int m;
      BufferedReader br= new BufferedReader(new InputStreamReader(System.in));
      String str;
      System.out.println("Enter month number: ");
      str= br.readLine();
      m= Integer.parseInt(str);
      try
      {
         if(m>12||m<1)
         throw new NumberFormatException();
         System.out.println("Month number entered is " + m);
      }
      catch(NumberFormatException ne)
      {
         System.out.println("Invalid month number! ");
      }
   }
}
```

## First Output:

```
Enter month number:
5
Month number entered is 5
```

## Second Output:

```
Enter month number:
35
Invalid month number!
```

# PRACTICAL 4A

**Aim: Write two Java programs to print numbers from 1 to 10 using:**

1. **By implementing the Runnable interface.**
2. **By extending the Thread class.**

**Input: By implementing the Runnable interface.**

```
class Numbers implements Runnable {
   public void run() {
      for (int i = 1; i <= 10; i++) {
         System.out.println(i);
      }
   }
}

public class Main {
   public static void main(String[] args) {
      Numbers n = new Numbers();
      Thread t1 = new Thread(n);
      t1.start();
   }
}
```

**Output:**

```
1
2
3
4
5
6
7
8
9
10
```

## Input: By extending the Thread class.

```
class Numbers extends Thread {
   public void run() {
      for (int i = 1; i <= 10; i++) {
         System.out.println(i);
      }
   }
}

public class Main {
   public static void main(String[] args) {
      Numbers n = new Numbers();
      Thread t1 = new Thread(n);
      t1.start();
   }
}
```

## Output:

```
1
2
3
4
5
6
7
8
9
10
```

# PRACTICAL 4B

**Aim: Write Java programs to print even and odd numbers from 1 to 50 using:**

1. **Runnable interface.**
2. **Extending the Thread class.**

**Input: By implementing the Runnable interface.**

```
class Numbers implements Runnable{
   public void run() {
      for (int i=1; i<=50; i++){
         if (i%2==0) {
            System.out.println (i + ":is an even number.");
         } else
         {
            System.out.println (i + ":is an odd number.");
         }
      }
   }
}

public class Main {
   public static void main(String[] args) {
      Numbers n= new Numbers();
      Thread t1= new Thread(n);
      t1.start();
   }
}
```

**Output:**

1:is an odd number.
2:is an even number.
3:is an odd number.
4:is an even number.
5:is an odd number.
6:is an even number.
7:is an odd number.
8:is an even number.
9:is an odd number.
10:is an even number.
11:is an odd number.
12:is an even number.
13:is an odd number.
14:is an even number.
15:is an odd number.
16:is an even number.
17:is an odd number.
18:is an even number.
19:is an odd number.
20:is an even number.
21:is an odd number.
22:is an even number.
23:is an odd number.
24:is an even number.
25:is an odd number.
26:is an even number.
27:is an odd number.
28:is an even number.
29:is an odd number.
30:is an even number.
31:is an odd number.
32:is an even number.
33:is an odd number.
34:is an even number.
35:is an odd number.
36:is an even number.
37:is an odd number.
38:is an even number.
39:is an odd number.
40:is an even number.
41:is an odd number.
42:is an even number.
43:is an odd number.
44:is an even number.
45:is an odd number.
46:is an even number.
47:is an odd number.
48:is an even number.
49:is an odd number.
50:is an even number.

**Input: By extending the Thread class.**

```
class Numbers extends Thread{
   public void run() {
      for (int i=1; i<=50; i++){
         if (i%2==0) {
            System.out.println (i + ":is an even number.");
         } else
         {
            System.out.println (i + ":is an odd number.");
         }
      }
   }
}

public class Main {
   public static void main(String[] args) {
      Numbers n= new Numbers();
      Thread t1= new Thread(n);
      t1.start();
   }
}
```

**Output:**

```
1:is an odd number.
2:is an even number.
3:is an odd number.
4:is an even number.
5:is an odd number.
6:is an even number.
7:is an odd number.
8:is an even number.
9:is an odd number.
10:is an even number.
11:is an odd number.
12:is an even number.
13:is an odd number.
14:is an even number.
15:is an odd number.
16:is an even number.
17:is an odd number.
18:is an even number.
19:is an odd number.
20:is an even number.
21:is an odd number.
22:is an even number.
23:is an odd number.
24:is an even number.
25:is an odd number.
26:is an even number.
27:is an odd number.
28:is an even number.
29:is an odd number.
30:is an even number.
31:is an odd number.
32:is an even number.
33:is an odd number.
34:is an even number.
35:is an odd number.
36:is an even number.
37:is an odd number.
38:is an even number.
39:is an odd number.
40:is an even number.
41:is an odd number.
42:is an even number.
43:is an odd number.
44:is an even number.
45:is an odd number.
46:is an even number.
47:is an odd number.
48:is an even number.
49:is an odd number.
50:is an even number.
```

# PRACTICAL 4C

**Aim: Write a Java Program to demonstrate isAlive() and join() method.**

**Input:**

```
class Alphabets extends Thread {
   public void run() {
      for (int i = 1; i <= 2; i++) {
         System.out.println((char)(i + 64));
         try {
            Thread.sleep(100);
         } catch (Exception e) {

         }
      }
   }
}

class Numbers extends Thread {
   public void run() {
      for (int i = 1; i <= 5; i++) {
         System.out.println(i);
         try {
            Thread.sleep(100);
         } catch (Exception e) {

         }
      }
   }
}

public class Main {
   public static void main(String[] args) {
      Alphabets n = new Alphabets();
      Thread t1 = new Thread(n);

      Numbers n1 = new Numbers();
```

```
        Thread t2 = new Thread(n1);

        t1.start();
        t2.start();

        System.out.println("Thread t1 is alive: " + t1.isAlive());
        System.out.println("Thread t2 is alive: " + t2.isAlive());

        try {
            t1.join();
            t2.join();
        } catch (InterruptedException ie) {

        }

        System.out.println("Thread t1 is alive: " + t1.isAlive());
        System.out.println("Thread t2 is alive: " + t2.isAlive());
    }
}
```

**Output:**

```
A
1
Thread t1 is alive: true
Thread t2 is alive: true
B
2
3
4
5
Thread t1 is alive: false
Thread t2 is alive: false
```

# PRACTICAL 5A

**Aim: Write a Java program to demonstrate the use of methods in linked list instance.**

**Input:**

```
import java.util.*;

class LinkedListDemo {
   public static void main(String ar[]) {
      LinkedList<String> list = new LinkedList<>();

      list.add("One");
      list.add("Two");
      list.add("Three");
      list.add("Four");
      list.add("Five");
      list.addLast("Six");
      list.addFirst("Seven");
      list.add(1, "Eight");

      System.out.println("Original Contents of list: " + list);

      list.remove("Five");
      list.remove(2);
      System.out.println("\nContents of list after deletion: " + list);

      list.removeFirst();
      list.removeLast();
      System.out.println("\nList after deleting first and last: " + list);

      String val = list.get(2);
      list.set(2, val+ "-Changed");
      System.out.println("\nList after change: " + list);
   }
```

**Output:**

Original Contents of list: [Seven, Eight, One, Two, Three, Four, Five, Six]

Contents of list after deletion: [Seven, Eight, Two, Three, Four, Six]

List after deleting first and last: [Eight, Two, Three, Four]

List after change: [Eight, Two, Three-Changed, Four]

# PRACTICAL 5B

**Aim: Write a Java program to demonstrate the simple use of class ArrayList.**

**Input:**

```
import java.util.*;
class ArrayListDemo {
   public static void main(String ar[]) {
      ArrayList<String> list = new ArrayList<>();
      System.out.println("\nInitilist size of list: "+list.size());

      list.add("First");
      list.add("Second");
      list.add("Three");
      list.add("Four");
      list.add("Five");
      list.add("Six");
      list.add(1, "Seven");

      System.out.println("\nSize of list after additions: " + list.size());

      System.out.println("\nContents of list: "+ list);

      list.remove("Five");
      list.remove(2);
      System.out.println("\nSize of list after deletions: " + list.size());

      System.out.println("\nContents of list: "+list);
   }
```

**Output:**

Initilist size of list: 0

Size of list after additions: 7

Contents of list: [First, Seven, Second, Three, Four, Five, Six]

Size of list after deletions: 5

Contents of list: [First, Seven, Three, Four, Six]

# PRACTICAL 5C

**Aim: Write a Java program to demonstrate the use of HashSet class and TreeSet class with many of their methods.**

**Input: HashSet class.**

```
import java.util.*;
class HashSetDemo
{
   public static void main(String ar[])
   {
      HashSet<String>hs=new HashSet<>();

      hs.add("one");
      hs.add("two");
      hs.add("three");
      hs.add("four");
      hs.add("five");
      hs.add("six");

      System.out.println(hs);
   }
}
```

**Output:**

[six, four, one, two, three, five]

**Input: TreeSet class with many of their methods.**

```
import java.util.*;
class TreeSetDemo
{
   public static void main(String ar[])
   {
      TreeSet<Integer>tree=new TreeSet<Integer>();

      tree.add(10);
      tree.add(22);
      tree.add(37);
      tree.add(49);

      Iterator iterator=tree.iterator();
      System.out.println("TreeSet data: ");
      while(iterator.hasNext())
      {
         System.out.print(iterator.next()+" ");
      }
      System.out.print("\n\nTree Set is empty: "+tree.isEmpty());
      System.out.println("\n\nFirst data: "+tree.first());
      System.out.println("Last data: "+tree.last());
      if(tree.remove(30))
      {
         System.out.println("Data is removed from TreeSet");
      }
      else
      {
         System.out.println("\nData doesn't exit!");
      }
      System.out.print("\nTreeSet contain: ");
      iterator=tree.iterator();
      while(iterator.hasNext())
      {
         System.out.println(iterator.next()+" ");
      }
      System.out.println("\nThe size of TreeSet: "+tree.size());
```

```
        tree.clear();
        System.out.println("First data: "+tree.isEmpty());
    }
}
```

**Output:**

TreeSet data:

10 22 37 49

Tree Set is empty: false

First data: 10

Last data: 49

Data doesn't exit!

TreeSet contain: 10

22

37

49

The size of TreeSet: 4

First data: true

# PRACTICAL 6

**Aim: Write a Java program to implement a Simple Registration Form using Java Swing.**

**Input:**

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

class MyFrame
   extends JFrame
   implements ActionListener {

   private Container c;
   private JLabel title;
   private JLabel name;
   private JTextField tname;
   private JLabel mno;
   private JTextField tmno;
   private JLabel gender;
   private JRadioButton male;
   private JRadioButton female;
   private ButtonGroup gengp;
   private JLabel dob;
   private JComboBox date;
   private JComboBox month;
   private JComboBox year;
   private JLabel add;
   private JTextArea tadd;
   private JCheckBox term;
   private JButton sub;
   private JButton reset;
   private JTextArea tout;
   private JLabel res;
   private JTextArea resadd;
```

```
private String dates[]
   = { "1", "2", "3", "4", "5",
      "6", "7", "8", "9", "10",
      "11", "12", "13", "14", "15",
      "16", "17", "18", "19", "20",
      "21", "22", "23", "24", "25",
      "26", "27", "28", "29", "30",
      "31" };
private String months[]
   = { "Jan", "feb", "Mar", "Apr",
      "May", "Jun", "July", "Aug",
      "Sup", "Oct", "Nov", "Dec" };
private String years[]
   = { "1995", "1996", "1997", "1998",
      "1999", "2000", "2001", "2002",
      "2003", "2004", "2005", "2006",
      "2007", "2008", "2009", "2010",
      "2011", "2012", "2013", "2014",
      "2015", "2016", "2017", "2018",
      "2019" };

public MyFrame()
{
   setTitle("Registration Form");
   setBounds(300, 90, 900, 600);
   setDefaultCloseOperation(EXIT_ON_CLOSE);
   setResizable(false);

   c = getContentPane();
   c.setLayout(null);

   title = new JLabel("Registration Form");
   title.setFont(new Font("Arial", Font.PLAIN, 30));
   title.setSize(300, 30);
   title.setLocation(300, 30);
   c.add(title);
```

```
name = new JLabel("Name");
name.setFont(new Font("Arial", Font.PLAIN, 20));
name.setSize(100, 20);
name.setLocation(100, 100);
c.add(name);

tname = new JTextField();
tname.setFont(new Font("Arial", Font.PLAIN, 15));
tname.setSize(190, 20);
tname.setLocation(200, 100);
c.add(tname);

mno = new JLabel("Mobile");
mno.setFont(new Font("Arial", Font.PLAIN, 20));
mno.setSize(100, 20);
mno.setLocation(100, 150);
c.add(mno);

tmno = new JTextField();
tmno.setFont(new Font("Arial", Font.PLAIN, 15));
tmno.setSize(150, 20);
tmno.setLocation(200, 150);
c.add(tmno);

gender = new JLabel("Gender");
gender.setFont(new Font("Arial", Font.PLAIN, 20));
gender.setSize(100, 20);
gender.setLocation(100, 200);
c.add(gender);

male = new JRadioButton("Male");
male.setFont(new Font("Arial", Font.PLAIN, 15));
male.setSelected(true);
male.setSize(75, 20);
male.setLocation(200, 200);
c.add(male);
```

```
female = new JRadioButton("Female");
female.setFont(new Font("Arial", Font.PLAIN, 15));
female.setSelected(false);
female.setSize(80, 20);
female.setLocation(275, 200);
c.add(female);

gengp = new ButtonGroup();
gengp.add(male);
gengp.add(female);

dob = new JLabel("DOB");
dob.setFont(new Font("Arial", Font.PLAIN, 20));
dob.setSize(100, 20);
dob.setLocation(100, 250);
c.add(dob);

date = new JComboBox(dates);
date.setFont(new Font("Arial", Font.PLAIN, 15));
date.setSize(50, 20);
date.setLocation(200, 250);
c.add(date);

month = new JComboBox(months);
month.setFont(new Font("Arial", Font.PLAIN, 15));
month.setSize(60, 20);
month.setLocation(250, 250);
c.add(month);

year = new JComboBox(years);
year.setFont(new Font("Arial", Font.PLAIN, 15));
year.setSize(60, 20);
year.setLocation(320, 250);
c.add(year);

add = new JLabel("Address");
add.setFont(new Font("Arial", Font.PLAIN, 20));
```

```
add.setSize(100, 20);
add.setLocation(100, 300);
c.add(add);

tadd = new JTextArea();
tadd.setFont(new Font("Arial", Font.PLAIN, 15));
tadd.setSize(200, 75);
tadd.setLocation(200, 300);
tadd.setLineWrap(true);
c.add(tadd);

term = new JCheckBox("Accept Terms And Conditions.");
term.setFont(new Font("Arial", Font.PLAIN, 15));
term.setSize(250, 20);
term.setLocation(150, 400);
c.add(term);

sub = new JButton("Submit");
sub.setFont(new Font("Arial", Font.PLAIN, 15));
sub.setSize(100, 20);
sub.setLocation(150, 450);
sub.addActionListener(this);
c.add(sub);

reset = new JButton("Reset");
reset.setFont(new Font("Arial", Font.PLAIN, 15));
reset.setSize(100, 20);
reset.setLocation(270, 450);
reset.addActionListener(this);
c.add(reset);

tout = new JTextArea();
tout.setFont(new Font("Arial", Font.PLAIN, 15));
tout.setSize(300, 400);
tout.setLocation(500, 100);
tout.setLineWrap(true);
tout.setEditable(false);
```

```
    c.add(tout);

    res = new JLabel("");
    res.setFont(new Font("Arial", Font.PLAIN, 20));
    res.setSize(500, 25);
    res.setLocation(100, 500);
    c.add(res);

    resadd = new JTextArea();
    resadd.setFont(new Font("Arial", Font.PLAIN, 15));
    resadd.setSize(200, 75);
    resadd.setLocation(580, 175);
    resadd.setLineWrap(true);
    c.add(resadd);

    setVisible(true);
  }

  public void actionPerformed(ActionEvent e)
  {
    if (e.getSource() == sub) {
      if (term.isSelected()) {
        String data1;
        String data
          = "Name : "
            + tname.getText() + "\n"
            + "Mobile : "
            + tmno.getText() + "\n";
        if (male.isSelected())
          data1 = "Gender : Male"
                + "\n";
        else
          data1 = "Gender : Female"
                + "\n";
        String data2
          = "DOB : "
            + (String)date.getSelectedItem()
```

```
                + "/" + (String)month.getSelectedItem()
                + "/" + (String)year.getSelectedItem()
                + "\n";

            String data3 = "Address : " + tadd.getText();
            tout.setText(data + data1 + data2 + data3);
            tout.setEditable(false);
            res.setText("Registration Successfully..");
        }
        else {
            tout.setText("");
            resadd.setText("");
            res.setText("Please accept the"
                    + " terms & conditions..");
        }
    }

    else if (e.getSource() == reset) {
        String def = "";
        tname.setText(def);
        tadd.setText(def);
        tmno.setText(def);
        res.setText(def);
        tout.setText(def);
        term.setSelected(false);
        date.setSelectedIndex(0);
        month.setSelectedIndex(0);
        year.setSelectedIndex(0);
        resadd.setText(def);
    }
  }
}

class Registration {

  public static void main(String[] args) throws Exception
  {
```

```
        MyFrame f = new MyFrame();
    }
}
```

**Output:**

# PRACTICAL 7

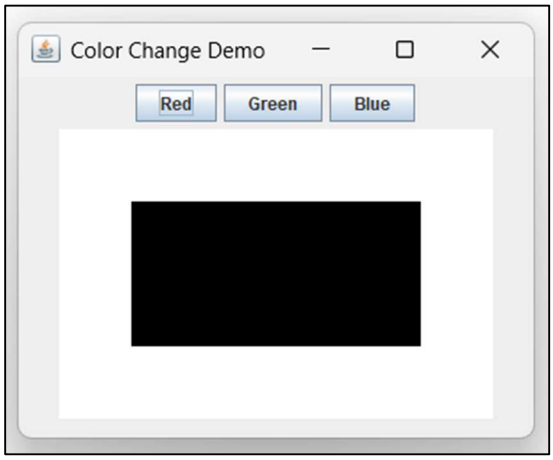
**Aim: Write a java program to fill rectangles with RED, GREEN, and BLUE colour according to the clicked button of specific colour.**

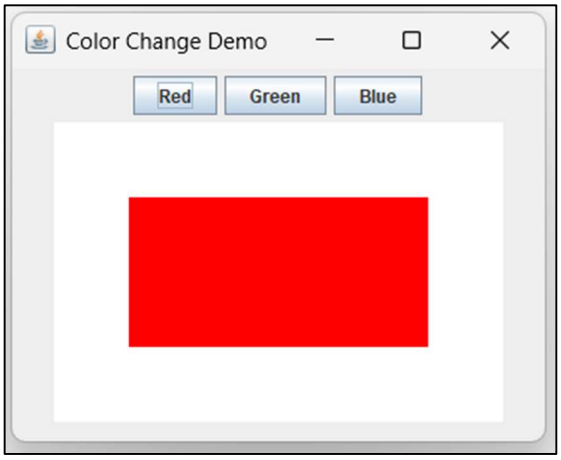
**Input:**

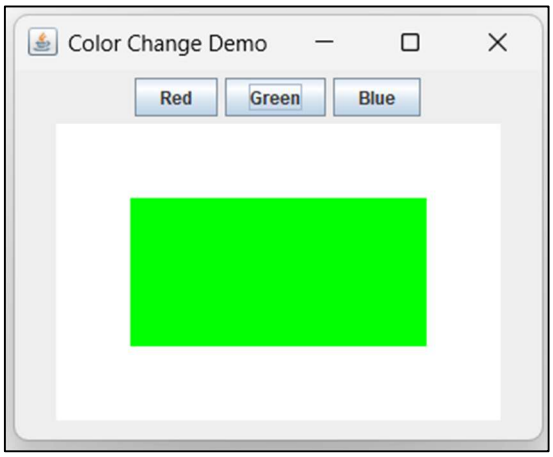
```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class EventDemo2Swing extends JFrame implements ActionListener {
   private Color currentColor = Color.BLACK;
   private JPanel drawPanel;

   public EventDemo2Swing() {
      setTitle("Color Change Demo");
      setSize(350, 350);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      setLayout(new FlowLayout());

      JButton redButton = new JButton("Red");
      JButton greenButton = new JButton("Green");
      JButton blueButton = new JButton("Blue");

      add(redButton);
      add(greenButton);
      add(blueButton);

      redButton.addActionListener(this);
      greenButton.addActionListener(this);
      blueButton.addActionListener(this);

      drawPanel = new JPanel() {
         @Override
         protected void paintComponent(Graphics g) {
            super.paintComponent(g);
            g.setColor(currentColor);
```

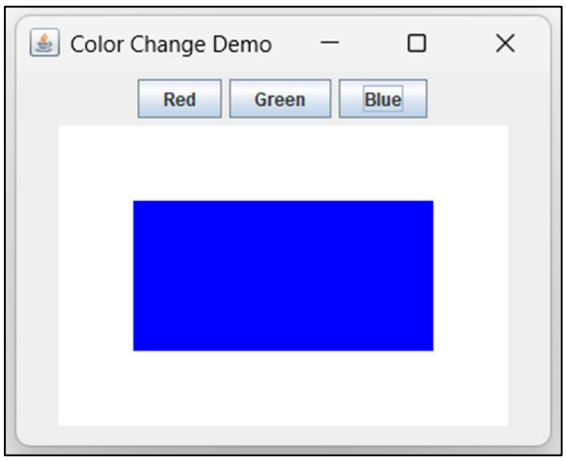
```
            g.fillRect(50, 50, 200, 100);
        }
    };

    drawPanel.setPreferredSize(new Dimension(300, 200));
    drawPanel.setBackground(Color.WHITE);
    add(drawPanel);

    setVisible(true);
  }

  public void actionPerformed(ActionEvent e) {
    String command = e.getActionCommand();

    switch (command) {
        case "Red":
            currentColor = Color.RED;
            break;
        case "Green":
            currentColor = Color.GREEN;
            break;
        case "Blue":
            currentColor = Color.BLUE;
            break;
    }

    drawPanel.repaint();
  }

  public static void main(String[] args) {
    new EventDemo2Swing();
  }
}
```

## Output:


Color Change Demo
Red
Green
Blue


Color Change Demo
Red
Green
Blue


Color Change Demo
Red
Green
Blue


Color Change Demo
Red
Green
Blue

# PRACTICAL 8

**Aim: Write a java program to insert one record of information into the database table named (dept table), by accepting the details from the user; using the Statement object.**

**Input:**

```
import java.sql.*;

import java.io.*;


class deptInsert {

   public static void main(String args[]) throws SQLException {

      try {

         Class.forName("com.mysql.jdbc.Driver");

         Connection con =
DriverManager.getConnection("jdbc:mysql://localhost/db", "root",
"Jadhav@123");

         Statement st = con.createStatement();


         BufferedReader br = new BufferedReader(new
InputStreamReader(System.in));


         System.out.print("Enter Dept. No.: ");

         int no = Integer.parseInt(br.readLine());

         System.out.print("Enter Dept. Name: ");

         String name = br.readLine();
```

```
            System.out.print("Enter Dept. Location: ");
            String loc = br.readLine();

            String sql1 = "INSERT INTO depttable VALUES(" + no + ", '" + name + "', '" +
loc + "')";
            st.executeUpdate(sql1);

            System.out.println("\nRecord Saved!!");
            con.close();

        } catch (Exception e) {
            System.out.println("Error1: " + e.getMessage());
        }
    }
}
```

## Output: From Frontend side

Enter Dept. No.: 1
Enter Dept. Name: E-MRD
Enter Dept. Location: Kamothe

Record Saved!!

Enter Dept. No.: 2
Enter Dept. Name: MRD
Enter Dept. Location: Kamothe

Record Saved!!

## Output: From Database end

| deptno | deptname | deptloc |
|---|---|---|
| NULL | NULL | NULL |

| deptno | deptname | deptloc |
|---|---|---|
| 1 | E-MRD | Kamothe |
| NULL | NULL | NULL |

| deptno | deptname | deptloc |
|---|---|---|
| 1 | E-MRD | Kamothe |
| 2 | MRD | Kamothe |
| NULL | NULL | NULL |

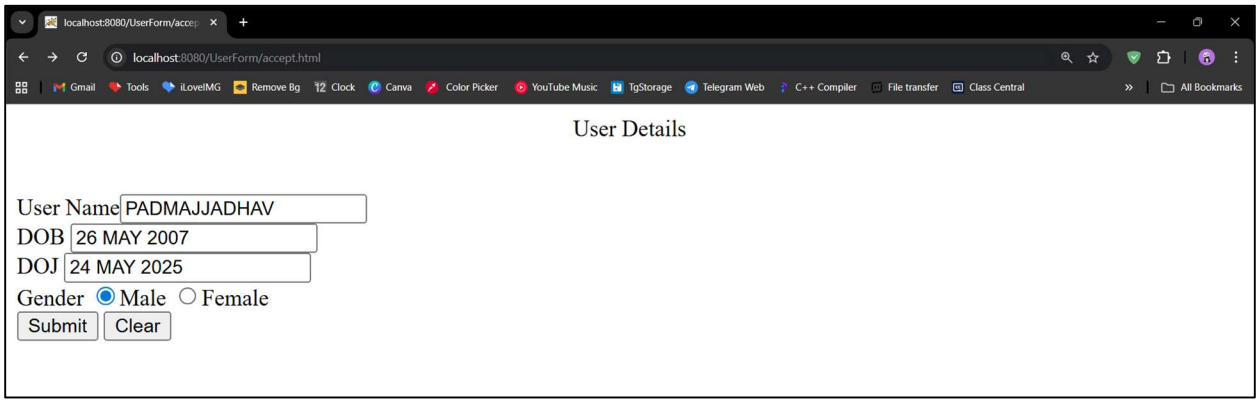
# PRACTICAL 9

**Aim: Write a java program for JSP code to print the details entered in the form on next page.**

**Input: accept.html**

```
<html>

<body>
   <form name=frm method=post action=display.jsp>
      <center> User Details</center>
      <br><br> User Name<input type=text name=t1>
      <br> DOB <input type=text name=t2>
      <br> DOJ <input type=text name=t3>
      <br> Gender
      <input type=radio name=r1 value="Male">Male
      <input type=radio name=r1 value="Female">Female
      <br><input type=submit value=submit>
      <input type=reset value=clear>
   </form>
</body>

</html>
```

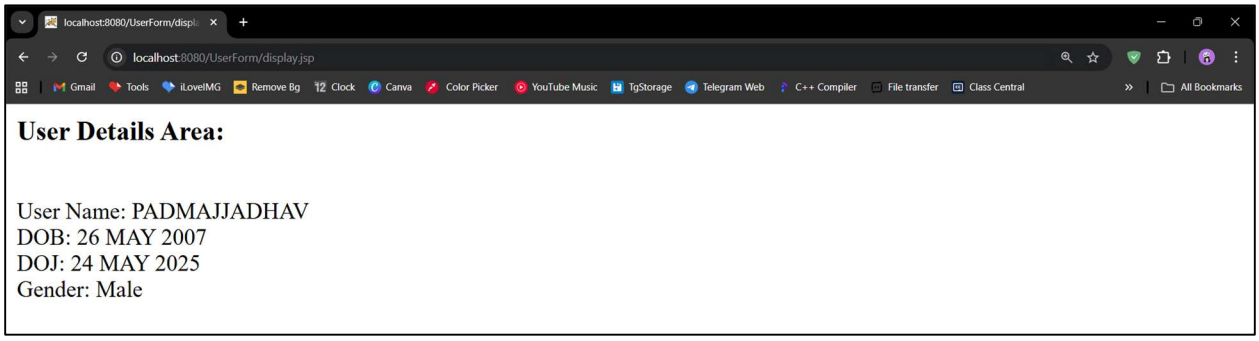
**Output:**



## Input: display.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%>
<html>
<body>
<%
   String username = request.getParameter("t1");
   String dob      = request.getParameter("t2");
   String doj      = request.getParameter("t3");
   String gender   = request.getParameter("r1");

   out.println("<h3>User Details Area:</h3>");
   out.println("<br>User Name: " + username);
   out.println("<br>DOB: " + dob);
   out.println("<br>DOJ: " + doj);
   out.println("<br>Gender: " + gender);
%>
</body>
</html>
```

## Output:



# PRACTICAL 10

**Aim: Write a java program for JavaScript file with JSON.**

**Input:**

```
<!DOCTYPE html>
<html>
<head>
   <title>First JSON example</title>
</head>
<body>
   <div id="content"></div>

   <script type="text/javascript">
      var object1 = { "name": "Mango", "rate": 250 };
      var object2 = { "name": "Apple", "rate": 150 };

      var content = "<h2>JSON with JavaScript example</h2><br>";

      content += "<h3>Fruit Name = " + object1.name + "</h3>";
      content += "<h3>Rate/Kg = " + object1.rate + "</h3><br>";

      content += "<h3>Fruit Name = " + object2.name + "</h3>";
      content += "<h3>Rate/Kg = " + object2.rate + "</h3><hr>";

      content += object1.name + " is available at " + object1.rate + " rate/kg<hr>";
      content += object2.name + " is available at " + object2.rate + " rate/kg<hr>";

      document.getElementById("content").innerHTML = content;
   </script>
</body>
</html>
```

**Output:**

# JSON with JavaScript example

**Fruit Name = Mango**

**Rate/Kg = 250**

**Fruit Name = Apple**

**Rate/Kg = 150**

---

Mango is available at 250 rate/kg

---

Apple is available at 150 rate/kg

---